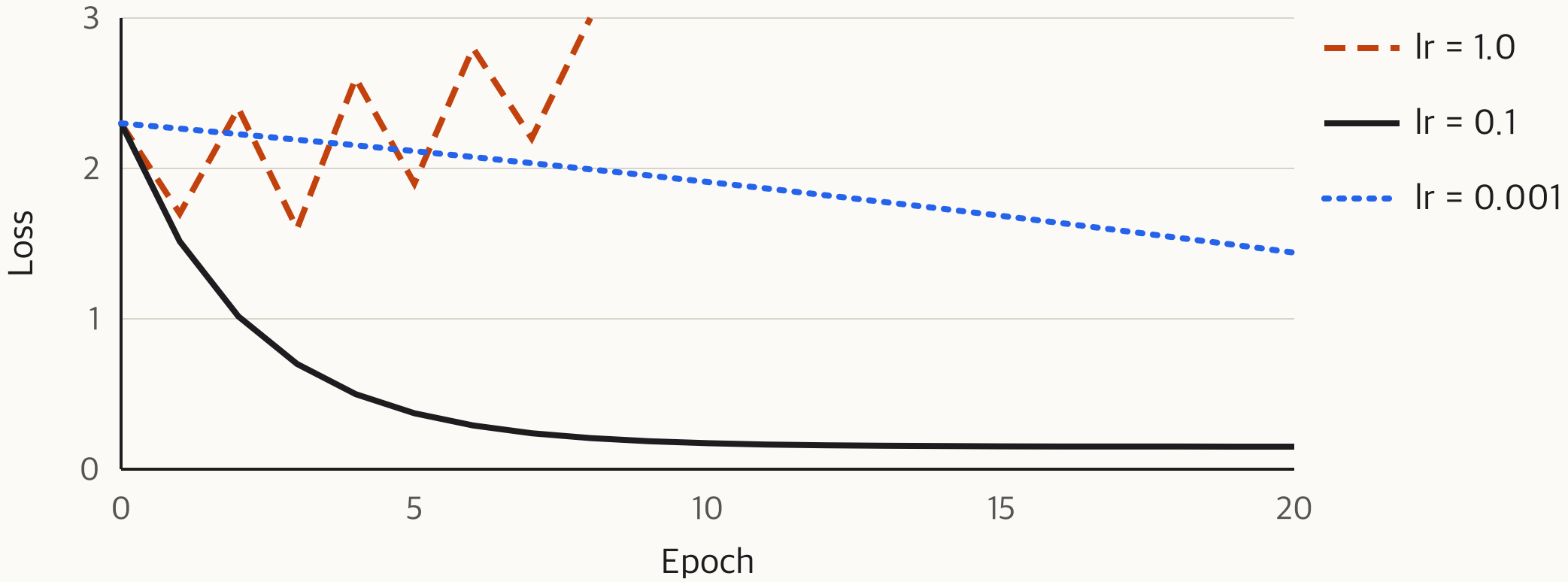
Gradient Descent & Learning Rate

Week 4

Gradient Descent

- Update weights in the direction that lowers the **loss function**
- **Learning rate**: step size of each update
- Too large → overshoots the minimum, loss may **diverge**
- Too small → slow training, may not **converge** within the epochs

Learning Rate Comparison



Stochastic Gradient Descent (SGD)

- Compute the gradient on a **mini-batch** instead of the full dataset
- Faster updates per step
- Gradient noise can help escape shallow **local minima**

Learning Rate Scheduling

- **Step decay**: halve the learning rate every few epochs
- **Cosine annealing**: decrease along a cosine curve
- **Warmup**: start small and increase the learning rate during the first 5 epochs

Next: Momentum, Adam optimizer